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Chapter 1

Pathological ECG Patterns: Clinical Descriptions and Reproduction

This chapter describes every pathological ECG pattern producible by the simulator, with clinical meaning and step-by-step reproduction instructions for each.

1.1 AV Conduction Abnormalities

1.1.1 First-Degree AV Block

Clinical Description

First-degree AV block is a **conduction delay** through the AV node, resulting in prolonged interval (> 200 ms), but with maintained 1:1 atrial-to-ventricular conduction. All P waves are followed by QRS complexes; no beats are dropped.

ECG Characteristics

- **PR interval:** Prolonged (typically 200–300 ms; normal is 120–200 ms)
- **QRS duration:** Normal (< 120 ms)
- **QRS morphology:** Normal (unless concurrent bundle branch block)
- **Heart rhythm:** Regular
- **P wave:** Normal morphology, slightly earlier than normal

Simulator Parameters: Reproduction

1. **Engine:** Use Phase 1/2 (faster, more stable pathology models)
2. **Pathology Panel:** Check “First-Degree AV Block”

3. **Parameter Adjustment** (manual, if needed):

- Navigate to **Parameters** panel → **AV Node** section
- **AV Delay**: Increase from default 100 ms to 150–200 ms
- Observe PR interval lengthening in real-time

4. **Expected Output**:

- P wave at normal time
- PR interval visibly longer than normal (e.g., 200 ms vs. 160 ms)
- QRS narrow and normal
- Repeats every cycle

Implementation

File: cardiac_sim/plugins/av_blocks.py

```
blackclass FirstDegreeAVBlock(AbstractPathologyPlugin):
    def name(self) -> str:
        return "First-Degree AV Block"

    def apply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
        self._original_av_delay = params.av_node.av_delay_ms
        params.av_node.av_delay_ms = 150.0 # Increase by 50 ms
        engine.apply_pathology(self)

    def unapply(self, engine: SimulationEngine, params: SimulationParameters) -> None:
        params.av_node.av_delay_ms = self._original_av_delay
        engine.apply_pathology(None)
```

1.1.2 **Second-Degree AV Block (Mobitz II)**

Clinical Description

Mobitz II AV block involves **occasional dropped beats**. Some atrial activations fail to conduct to the ventricles; these manifest as P waves without following QRS complexes.

Unlike Mobitz I (progressive PR prolongation), Mobitz II has:

- Constant interval when conduction occurs
- Suddenly dropped beat(s) with no prior warning
- Often indicates infra-nodal conduction disease (His-Purkinje system)
- More dangerous (risk of complete heart block)

ECG Characteristics

- **PR interval:** Constant (normal or mildly prolonged)
- **Dropped beats:** Periodic (e.g., 2:1, 3:1 conduction ratios)
- **Appearance:** P wave without QRS; irregular rhythm
- **QRS duration:** May be widened (if His-Purkinje disease)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2
2. **Pathology Panel:** Check “Second-Degree AV Block (Mobitz II)”
3. **Alternative Manual Method:**
 - **Parameters** → **AV Node** section
 - Set **AV Conductance** = 0.5 (50% conduction; some beats blocked)
 - May need to adjust **AV Refractory Period** to 400 ms
4. **Expected Output:**
 - Regular P wave rhythm (atrial activity unchanged)
 - Every 2nd, 3rd, or variable P wave followed by QRS
 - Dropped beats appearing as P waves alone

1.1.3 Third-Degree AV Block (Complete Heart Block)

Clinical Description

Complete AV block is **total failure of AV conduction**. Atria and ventricles beat independently (“AV dissociation”):

- Atria: Normal sinus rhythm, 60–80 bpm
- Ventricles: Escape rhythm (junctional or idioventricular) at much slower rate, 30–40 bpm

ECG Characteristics

- **P waves:** Regular, independent of QRS; may appear anywhere in the ECG
- **QRS complexes:** Slow rate (30–60 bpm), wide morphology
- **PR relationship:** Completely variable (sometimes short, sometimes P in QRS, sometimes after QRS)
- **Axis:** May show left axis deviation or extreme axis (if escape rhythm from LV)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (recommended; cleaner demonstration)
2. **Pathology Panel:** Check “Third-Degree AV Block (Complete)”
3. **Or Manual Method:**
 - **Parameters** → **AV Node**
 - Set **AV Conductance** = 0.0 (complete block)
 - Set **Ventricular** → **Ectopic Trigger Rate** to 40 bpm (escape rhythm)
4. **Expected Output:**
 - Atrial rhythm regular at normal rate (70 bpm)
 - Ventricular rhythm regular at escape rate (40 bpm)
 - P waves and QRS complexes dissociated
 - QRS wide (>120 ms) if escape from ventricular tissue

1.2 Bundle Branch Blocks

1.2.1 Right Bundle Branch Block (RBBB)

Clinical Description

Conduction block in the **right bundle branch** (RBB) forces RV activation to occur via **collateral conduction through the left ventricle**. This delays RV activation, causing:

- Widened (>120 ms)
- Characteristic **terminal R’ wave** (second positive deflection) in lead V1
- S wave in leads I and V5-V6

ECG Characteristics

- **QRS duration:** ≥ 120 ms (often 120–150 ms)
- **Lead V1:** Characteristic “M” or “rSR’” pattern (small r, deep S, tall R’)
- **Leads I, V5-V6:** S wave (negative deflection)
- **PR interval:** Normal
- **Axis:** Normal or rightward

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Right Bundle Branch Block”
3. **Manual Method:**
 - **Parameters** $\rightarrow$ **Ventricular**
 - Set **RBB Conductance** = 0.0
 - Observe V1 lead: R’ should become visible
4. **Expected Timing** (Phase 3):
 - RV activation delayed to ≈ 645 ms (vs. ≈ 435 ms normal)
 - QRS broadened to ≈ 120 ms (vs. ≈ 70 ms)
 - T wave morphology also affected (opposite direction)

1.2.2 Left Bundle Branch Block (LBBB)

Clinical Description

Block in the **left bundle branch** delays LV activation, forcing it to propagate from RV via collaterals. This causes:

- Widened (>120 ms, often 140–160 ms)
- Broad, notched positive deflection in lateral leads (I, aVL, V5-V6)
- Absent Q wave in lateral/inferior leads (activation from RV)
- Characteristic “W” pattern in lead V1

ECG Characteristics

- **QRS duration:** ≥ 120 ms (typically 140–160 ms)
- **Leads I, V5-V6:** Broad positive (R) wave with notch or split
- **Lead V1:** W pattern (rS or QS)
- **Absent Q waves:** In leads I, aVL, V5-V6 (no normal Q wave in these leads)
- **Axis:** Often leftward (left axis deviation)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Left Bundle Branch Block”
3. **Manual Method:**
 - **Parameters** $\rightarrow$ **Ventricular**
 - Set **LBB Conductance** = 0.0
 - Observe lateral leads: broadened R wave with notch
4. **Expected Timing** (Phase 3):
 - LV activation delayed to ≈ 645 ms
 - QRS very broad (≈ 150 ms)
 - LV dominates morphology (rightward dipole from delayed LV)

1.3 Atrial Arrhythmias

1.3.1 Atrial Fibrillation (AFib)

Clinical Description

Atrial fibrillation is chaotic, uncoordinated electrical activity in the atria. The atrial rate exceeds 300 bpm with no organized conduction. Only occasional atrial activations penetrate the AV node, producing:

- Absent P waves (replaced by erratic baseline “fibrillatory waves”)
- Irregularly irregular ventricular rhythm
- Narrow complexes (if conduction intact below AV node)

ECG Characteristics

- **P waves:** Absent; baseline replaced by fine or coarse undulations (f waves)
- **Baseline:** Erratic, no isoelectric line
- **Ventricular rate:** Irregularly irregular (e.g., 60–100 bpm in controlled AFib)
- **QRS:** Narrow (<120 ms) unless concurrent BBB
- **RR intervals:** Variable (no two RR intervals the same)

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (Phase 3 would require re-entrant circuit modeling)
2. **Pathology Panel:** Check “Atrial Fibrillation”
3. Implementation detail:
 - Atrial activation is replaced by chaotic re-entrant patterns
 - AV node conduction becomes probabilistic (some atrial waves blocked)
 - Baseline shows characteristic fibrillatory waves
4. **Expected Output:**
 - No P waves visible
 - Baseline wiggles (fine or coarse f waves)
 - QRS complexes at irregular intervals

1.3.2 Sinus Bradycardia

Clinical Description

Sinus bradycardia is a normal sinus rhythm at a rate < 60 bpm. All intervals and morphologies are normal; only the rate is slow.

Causes include athletic conditioning, hypothyroidism, increased vagal tone, or certain medications.

ECG Characteristics

- **Heart rate:** < 60 bpm (e.g., 50 bpm)
- **PR interval:** Normal
- **QRS:** Normal

- **Rhythm:** Regular
- **P-QRS-T morphology:** Completely normal; unchanged from NSR

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Sinus Bradycardia”
3. **Manual Method:**
 - **Parameters** → **SA Node**
 - Set **Cycle Length (ms)** = 1200 (corresponds to 50 bpm)
 - Formula: $\text{HR (bpm)} = 60,000 / \text{Cycle Length (ms)}$
 - So 1200 ms → 50 bpm
4. **Expected Output:**
 - All complexes identical to normal
 - HR display shows 50 bpm
 - Horizontal spacing between complexes larger than normal

1.3.3 Sinus Tachycardia

Clinical Description

Sinus tachycardia is normal sinus rhythm at a rate > 100 bpm. All intervals and morphologies remain normal; only the rate is fast.

Causes include exercise, fever, anemia, hyperthyroidism, or sympathetic stimulation.

ECG Characteristics

- **Heart rate:** > 100 bpm (e.g., 140 bpm)
- **PR interval:** Normal
- **QRS:** Normal
- **Rhythm:** Regular
- **P wave:** May be peaked (increased amplitude) due to faster rate
- **T wave:** May slightly overlap preceding P wave

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 or Phase 3
2. **Pathology Panel:** Check “Sinus Tachycardia”
3. **Manual Method:**
 - **Parameters** → **SA Node**
 - Set **Cycle Length (ms)** = 400 (→ 150 bpm)
 - Formula: $60,000 / 400 = 150$ bpm
4. **Expected Output:**
 - All complexes identical to normal
 - HR display shows 150 bpm
 - Horizontal spacing between complexes narrower than normal
 - T wave may slightly overlap next P wave

1.4 Ventricular Arrhythmias

1.4.1 Premature Ventricular Contraction (PVC)

Clinical Description

A **PVC** is an ectopic ventricular beat that occurs earlier than expected in the cardiac cycle. On the ECG:

- Wide (>120 ms) with abnormal morphology
- Abnormal T wave (opposite polarity to QRS)
- Compensatory pause (next sinus beat occurs at normal time)
- Variable coupling interval (time from previous QRS to PVC)

ECG Characteristics

- **QRS duration:** >120 ms, often 140–180 ms
- **QRS morphology:** Abnormal; often broad and slurred
- **T wave:** Opposite polarity to QRS (discordant ST-T)
- **PR/PP interval:** No P wave before the PVC
- **Coupling interval:** Relatively fixed for multiple PVCs of same origin

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (preferred) or Phase 3
2. **Pathology Panel:** Check “Premature Ventricular Contraction”
3. **Manual Method:**
 - **Parameters** → **Ventricular**
 - Set **Ectopic Trigger Rate (bpm)** = 60
 - Set **Ectopic Coupling Interval (ms)** = 600 (PVC fires 600 ms after normal beat)
4. **Expected Output:**
 - Normal sinus beats
 - Every so often, wide QRS with abnormal morphology
 - Compensatory pause after PVC (total RR interval = $2 \times$ sinus interval)

1.4.2 Ventricular Fibrillation (VF)

Clinical Description

Ventricular fibrillation is chaotic, uncoordinated electrical activity in the ventricles. There is no organized contractile function; the ECG shows:

- Erratic, low-amplitude oscillations (“fine VF”) or larger waves (“coarse VF”)
- No identifiable QRS, T, or ST segment
- Hemodynamic collapse (loss of perfusion)
- Lethal rhythm unless rapidly defibrillated

ECG Characteristics

- **Baseline:** Irregular undulations, no isoelectric segments
- **Amplitude:** Fine VF shows low amplitude (<1 mm); coarse VF shows >5 mm waves
- **Frequency:** 150–300 Hz baseline oscillations
- **QRS/T:** Absent
- **Rate:** Unmeasurable

Simulator Parameters: Reproduction

1. **Engine:** Phase 1/2 (with special re-entrant modeling) or Phase 3
2. **Pathology Panel:** Check “Ventricular Fibrillation”
3. Implementation detail:
 - Ventricular conduction is randomized
 - Multiple competing depolarization foci activate ventricles chaotically
4. **Expected Output:**
 - Erratic baseline; no organized complexes
 - Fine or coarse undulations depending on rate of ectopic foci

Chapter 2

Architecture and Design Diagrams

2.1 Class Hierarchy Diagram

[Diagram - text description in document]

Figure 2.1: Class hierarchy and composition relationships. Abstract `SimulationEngine` is implemented by two concrete engines. `ConductionCableEngine` owns a `ConductionCable1D` and uses `LeadFieldModel`. GUI components interface with a worker thread.

2.2 Function Call Graph (Phase 3)

[Diagram - text description in document]

Figure 2.2: Main simulation loop call graph for Phase 3 engine. Each frame (~ 2 ms), `step` is called, which advances the cable via 20 micro-steps and computes the ECG dipole.

2.3 Data Flow: Parameter Changes

[Diagram - text description in document]

Figure 2.3: Data flow when user adjusts a parameter. The change propagates through the parameter model to the engine, which reinitializes and begins generating new ECG samples reflected in the display.

2.4 State Machine: Beat Lifecycle (Phase 3)

[Diagram - text description in document]

Figure 2.4: Beat state machine for `ConductionCableEngine`. A beat transitions from “No Beat” to “In Progress” when the SA node fires, then to “Committed” when all ventricular nodes have activated or a timeout is reached.

Chapter 3

Technical Reference and Appendices

3.1 FHN Model Constants and Ranges

Parameter	Phase 3-D Value	Notes
τ (time scale)	15 ms	Calibrated for correct P/PR/QRS timing
ε (recovery rate)	0.08	Controls APD; smaller = longer APD
a (FHN param)	0.7	Typical literature value
b (FHN param)	0.8	Typical literature value
Δt_{int} (micro-step)	0.1 ms	10 steps per output frame
N_{atr} (atrial cells)	8	SA + RA + LA
N_{ven} (ventricular cells)	10	His + LBB + LV
$D_{\text{atr,max}}$	2.88	Atrial diffusion
$D_{\text{ven,max}}$	72.0	Ventricular diffusion (His-LBB pathway)
D_{lv}	3.0	LV myocardial diffusion
V_{rest}	-1.2	Normalized rest potential
V_{peak}	+2.0	Normalized depolarization peak
$V_{\text{threshold}}$	-0.5	Activation threshold (normalized)

Table 3.1: FHN model constants for Phase 3-D (final calibration).

3.2 ECG Node Configuration (Phase 3)

Node	Depol σ (ms)	Repol Delay (ms)	Repol σ (ms)	Role
SA	12.7	280	50	SA node
RA	12.7	280	50	Right atrium
LA	12.7	280	50	Left atrium
AV	12.7	280	50	AV node
HIS	12.7	280	50	His bundle
LBB	12.7	280	50	L. bundle branch
RBB	12.7	280	50	R. bundle branch
SEPT	12.7	280	50	Septum
LV	40.0	280	50	L. ventricle (widened depol)
RV	12.7	280	50	R. ventricle

Table 3.2: ECG node Gaussian parameters. LV depol_duration widened to 40 ms so adjacent nodes produce overlapping Gaussians $\rightarrow$ smooth QRS morphology.

3.3 Installation and Running

3.3.1 Prerequisites

- Python 3.12+
- PyQt6 6.0
- NumPy 2.4 (Numba requires NumPy $\leq$ 2.4; we use 2.5 which forces fallback)
- pyqtgraph 0.13.0
- (Optional) Numba 0.59 for JIT compilation (requires compatible NumPy)

3.3.2 Setup Steps

```

black# Clone repository
cd ECG_simulator

# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate # or .venv\textbackslashScripts\textbackslashactivate

# Install dependencies
pip install -r requirements.txt

```

```
# Run application
python main.py
```

3.3.3 Running Tests

```
black# Run all unit tests

# Run specific test file
python -m pytest tests/test_cable_model.py -v

# Run with coverage report
python -m pytest tests/ --cov=cardiac_sim --cov-report=html
```

pyth

3.4 Performance Considerations

3.4.1 Phase 1/2 (Graph Engine)

- Execution time per step: ≈ 0.1 ms
- Bottleneck: BFS graph traversal and Gaussian dipole summation
- Scales well to 1000+ Hz

3.4.2 Phase 3 (Cable Engine)

- Execution time per step: $\approx 1\text{--}2$ ms (on modern CPU)
- Simulated time per wall-clock second: $\approx 2.2\times$ real-time
- Bottleneck: $N \times M$ FHN integration (cables + micro-steps)
- Numba JIT (if available): $\approx 10\text{--}20\times$ speedup possible
- Current: NumPy fallback due to NumPy 2.5 incompatibility

3.5 Validation Results

3.5.1 Phase 3-D Timing Validation

Interval	Simulator	Clinical Target
P wave duration	64 ms	40–100 ms
PR interval	196 ms	120–200 ms
QRS duration	70 ms	<120 ms
QT interval	360 ms	300–440 ms
Heart rate	70 bpm	60–100 bpm

Table 3.3: Phase 3-D validated timing intervals compared to clinical targets (normal sinus rhythm at 70 bpm).

3.5.2 Multi-Beat Rhythm (Phase 3)

Successfully generates 6 consecutive QRS complexes at 70 bpm over 5 seconds without numerical artifacts or premature termination.

3.5.3 Amplitude Validation (Phase 3)

- QRS peak-to-trough: ≈ 1.3 mV (clinical: 0.5–2.5 mV)
- T wave amplitude: ≈ 0.6 mV
- QRS/T ratio: ≈ 2.2 (clinical normal: 1–2.5)

3.6 File Organization Summary

File Path	Purpose	Lines
main.py	Application entry point	50
cardiac_sim/core/parameter_model.py	Parameter dataclasses	200
cardiac_sim/core/interfaces.py	Abstract base classes	80
cardiac_sim/core/tissue/cable_1d.py	FHN cable implementation	500
cardiac_sim/core/conduction/graph.py	Graph structure	150
cardiac_sim/core/cell_models/fitzhugh_nagumo.py	FHN cell model	100
cardiac_sim/core/ecg/lead_field.py	Lead vectors	120
cardiac_sim/core/simulation_worker.py	Threading wrapper	150
cardiac_sim/simulation/graph_engine.py	Phase 1/2 engine	400
cardiac_sim/simulation/cable_engine.py	Phase 3 engine	450

File Path	Purpose	Lines
cardiac_sim/gui/main_window.py	Main GUI window	250
cardiac_sim/gui/parameter_panel.py	Parameter tree UI	180
cardiac_sim/gui/pathology_panel.py	Pathology selector	150
cardiac_sim/gui/ecg_display.py	ECG plotter	200
cardiac_sim/plugins/av_blocks.py	AV block pathologies	100
cardiac_sim/plugins/bundle_blocks.py	BBB pathologies	80
cardiac_sim/plugins/atrial.py	Atrial pathologies	100
cardiac_sim/plugins/ventricular.py	Ventricular pathologies	120
cardiac_sim/plugins/sinus_rhythms.py	Sinus rhythm variants	80
Total		≈ 3500

3.7 Glossary of Abbreviations

Abbreviation	Meaning
SA	Sinoatrial (atrial pacemaker)
AV	Atrioventricular (AV node)
RA/LA	Right/Left atrium
RV/LV	Right/Left ventricle
RBBB/LBBB	Right/Left bundle branch block
PR	Atrial to ventricular conduction time
QRS	Ventricular depolarization duration
ST	Repolarization segment
ECG	Electrocardiogram
APD	Action potential duration
CV	Conduction velocity
FHN	FitzHugh-Nagumo (ionic model)
PVC	Premature ventricular contraction
AFib	Atrial fibrillation
VF	Ventricular fibrillation

3.8 References and Further Reading

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4. Roth, B. J. (1997). Electrical conductivity values used with the bidomain model of cardiac tissue. *IEEE Trans. Biomed. Eng.*, 44(4), 326–328.
5. Maringhini, G., et al. (2015). ECG simulation using cellular automata. *Comput. Biol. Med.*, 65, 207–215.